PUBLIC WINDOWS EDITION

LRC BULK FINDER

COMPLETE USER MANUAL

Find synchronized lyrics for your MP3 library, prepare Greek titles with optional local AI, and export safe .lrc sidecar files.

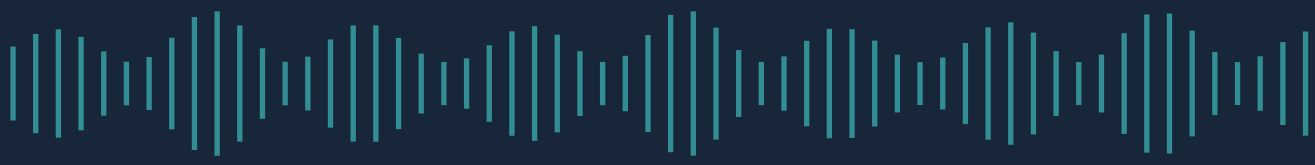
STEP-BY-STEP SETUP • GREEK + INTERNATIONAL MUSIC

FAST/FULL MODES • OFFLINE AI SETUP • REPORTS

SMALL TEST MODE • OVERNIGHT RUNS • SAFETY

Version 2.1.4 | Python 3.10+ | Optional AUEB G2G AI

Created by Dimitris L.



GENERAL-USE RELEASE • OCTOBER 2026

01 Install and start

This general-use package contains **no private song list**, no bundled MP3s, and no pre-generated lyrics. It starts with an empty manual-alias file. The program works with a folder of MP3s on your own Windows computer.

What you need

Requirement	Details
Windows PC	Windows 10/11; internet for lyrics provider searches.
Python	Python 3.10 or later from python.org; 3.13 is a tested configuration.
MP3 collection	Files with Artist / Title ID3 tags are strongly recommended.
Optional Greek AI	gr-nlp-toolkit with a large local G2G model (see next page).

First-time installation

1. Download and **Extract All** from the ZIP into its own folder. Do not run directly inside the ZIP.

2. Install Python if needed. Open PowerShell and verify it:

```
py --version
```

3. Double-click **START_HERE.cmd** or open PowerShell in the extracted folder and run:

```
.\START_HERE.cmd
```

4. For LRCLIB support, add your **real project HTTPS URL or contact email** to **LRCLIB_CONTACT.txt**. Without it, LRCLIB is skipped; the other sources remain available.

5. The launcher checks core Python packages, then asks for your music folder and search settings.

Important
The program contacts external lyric services with artist/title metadata, but does not upload MP3 audio. It creates missing **.lrc** sidecars and reports; it does not edit music files.

02 Optional Greeklish AI setup

The AUEB **gr-nlp-toolkit** model converts likely Greeklish into Greek-script search alternatives. It is useful for Greek music, but not needed for non-Greek music. The AI model is **not included** in this ZIP.

Easy method: installer included

1. Run **SETUP_GREEK_AI.cmd** in the extracted folder.
2. Confirm Y. The script installs **gr-nlp-toolkit**, initializes the G2G model, and prints a test result.
3. The first model initialization may download approximately **1.2 GB** plus Python dependencies. Later runs normally reuse the locally cached model.

Manual method: exact PowerShell commands

```
py --version
py -m pip install gr-nlp-toolkit
```

Then run the following on one line in PowerShell:

```
py -c "from gr_nlp_toolkit import Pipeline; print(Pipeline('g2g')('Kalimera').text)"
```

Successful test
You should see text written with Greek characters. The spelling/accents can vary; the test simply proves the model loaded and performed inference.

When the downloader asks

Option	Choose
Greeklish / Greek folder?	Y to use Greek alternatives; otherwise N.
Use installed LOCAL G2G AI?	Y if setup succeeded. N uses original metadata + aliases.
Extra MusicBrainz lookup?	Start with N. It adds network metadata queries, not LRC data.

The model loads only when needed and its conversion results are cached locally as **G2G_AI_Cache.json**. It is a spelling/search aid, not a source of verified music releases.

03 Choose your search settings

All questions are asked **before conversion and searching**. You can choose a 20-song preview or run the entire remaining folder unattended.

Startup question	Greek / Greeklish	International music
Greeklish / Greek folder?	Y	N
Use local G2G AI?	Y after setup	Not asked
MusicBrainz metadata?	N initially	Not asked
Search mode?	FAST test, then FULL	FAST test, then FULL
Include subfolders?	Y if desired	Y if desired
How many missing songs?	20–30 test; A for all	20–30 test; A for all
Search automatically after preview?	N = preview only; Y = search	Not asked

FAST vs FULL

Mode	How it works	Use it for
FAST	5 prioritized routes: SyncLRC, NetEase, Unison, LrcAPI, LRCLIB.	Quick first pass and sample testing.
FULL	All 12 available routes, including some aggregators and experimental services.	Trying additional sources on songs that still lack LRC.

Efficient workflow
Run **FAST first**. Then run **FULL on the same folder** to look for the remaining songs; existing .lrc files are automatically skipped. FULL can take much longer and may not find anything extra.

04 Test safely before a large run

For a first run, choose **20** songs and check the results. Do not begin an all-night, thousand-song job before confirming file naming and lyric accuracy.

Greek sample with preview-only mode

1. Choose Greek mode **Y**, AI **Y**, MusicBrainz **N**.
2. Choose **FAST**, the appropriate subfolder setting, and **20** songs.
3. Choose **N** for “automatically search after preview.” The program creates CSV previews but makes **no lyrics-provider requests**.
4. Open **LRC_G2G_AI_Review_v2_1_4.csv**. Compare original and converted artist/title side by side.
5. Fix clear mistakes using **Greek_Aliases.csv**, then rerun with **Y** to search the same sample.

International music sample

1. Choose Greek mode **N** and FAST.
2. Select a representative folder with **20** songs.
3. Check the saved .lrc files against playback before selecting **A** for a large collection.

What a correct LRC result looks like

```
Example Artist - Example Song.mp3
Example Artist - Example Song.lrc

[00:12.30]First synchronized lyric line
[00:17.90]Second synchronized lyric line
```

Accuracy first
Even a close track duration does not guarantee the right recording or correctly aligned timestamps. Remixes, radio edits and live versions may differ. Unsure candidates should go to review instead of being automatically saved.

05 Greek aliases and AI accuracy

The package intentionally ships with an **empty Greek_Aliases.csv**. Your own corrected entries override AI suggestions and do not require changing the MP3 tags.

CSV format

```
type,greeklish,greek
title,Kalimera,Καλημέρα
artist,Eleni Papadopoulou,Ελένη Παπαδοπούλου
```

These are **format illustrations only**, not verified music-database entries. Open **Greek_Aliases.csv** in a spreadsheet or text editor, leave the header unchanged, and add a row for each confirmed correction. Save as CSV UTF-8.

How the Greek search works

Priority	What is searched
1 · Original	Original ID3 artist and original song title.
2 · Manual correction	Verified title/artist spellings from your Greek_Aliases.csv.
3 · AI candidate	Optional local model suggestions for likely Greeklish names.
4 · MusicBrainz	Optional metadata variations when enabled and available.

AI is not an authority
Greeklish spellings can be ambiguous. AI results may lack accents, choose the wrong word, or mis-handle version labels. Review the preview CSV and never assume a converted title identifies the correct recording.

Mixed Greek/Greeklish titles are supported. Clearly international English titles are usually not converted, though automatic detection can make mistakes. The program always retains the original metadata as a search option.

06 Search, retries and safety

The program contacts multiple sources but does not repeatedly search every song when one provider fails. The rules below are important for avoiding wasted time and service rate limits.

Provider result	Does Round 2 / 3 retry it?
Synchronized lyrics saved	No
Completed NO_MATCH / NOT_FOUND	No
Uncertain / rejected candidate	No automatic retry; review it
Skipped source / unavailable	No
HTTP 429 (rate-limited)	No; provider cooldown is remembered across restarts
HTTP 5xx or other server error	No automatic retry
Actual network read/connection timeout	Yes, only that exact song + provider

What is protected

- Your MP3 audio files and ID3 tags are never renamed, moved, edited, overwritten or deleted.
- Any MP3 with an existing matching **.lrc** file is skipped.
- Songs containing standalone **LIVE** in filename or ID3 title are excluded.
- Automatic saving requires resolved artist/title, same recording version, and an MP3-versus-provider duration gap of 3 seconds or less.
- Source APIs may be unavailable, change behavior, or throttle your connection.

Why “429” appears
HTTP 429 means a provider is rate-limiting requests. It does not necessarily mean your entire internet connection is blocked. The downloader records Retry-After in LRC_Provider_Cooldowns.json and avoids that provider until the cooldown expires.

07 Run unattended overnight

Once you have tested a small sample, the program can run without further configuration questions. **All decisions are made before the AI preview or online searches begin.**

Recommended overnight checklist

- **Keep the computer awake:** temporarily adjust Windows power/sleep settings. You can turn off only the display.
- **Use a stable connection** and avoid running multiple large lyric jobs simultaneously.
- **Close no windows:** leave the Command Prompt running. The final Enter prompt appears only after results are written.
- **Greek folders:** Y for Greek mode, Y for G2G if installed, N for MusicBrainz initially, FAST or FULL, A for all, Y to auto-search.
- **International folders:** N for Greek mode, FAST initially, A for all; later rerun FULL for missing lyrics.

Expected progress stages

Stage	What is happening
Scan	Find MP3s, exclude LIVE, skip existing .lrc files.
Greek preflight	Generate cached Greek AI candidates and save CSV previews.
Round 1	Search configured providers for each eligible track.
Rounds 2–3	Retry only actual timeouts on exact song/provider pairs.
Finish	Save Not Found, Timeouts, reviews, and statistics CSVs.

Performance expectations
FULL mode can take hours for a large library: provider calls may be slow, and some have limited availability. The number of successful LRC downloads is never guaranteed.

08 Understand the output files

Reports are created in the **selected music folder**. The AI cache is kept beside the program so repeated conversions can be reused on future runs.

Output filename	What it tells you
LRC_Not_Found.txt	Songs with no accepted synced lyrics from completed searches; coverage may be partial.
LRC_Timeouts.txt	Songs/searches unfinished by timeout, rate limit, or server error.
LRC_Review_Needed.txt	Songs with uncertain lyric candidates: a separate human review list.
LRC_Finder_Report_v2_1_4.csv	Per-track outcome, attempted searches and success details.
LRC_Finder_Review_v2_1_4.csv	Rejected candidates and paths to available review-only lyric text files.
LRC_Finder_Provider_Details_v2_1_4.csv	Which source returned NO_MATCH, was blocked or supplied lyrics.
LRC_Source_Statistics_v2_1_4.csv	Aggregated result counts by source.
LRC_Greek_Search_Preview_v2_1_4.csv	Greek title alternatives and original metadata.
LRC_G2G_AI_Review_v2_1_4.csv	Side-by-side AI conversion, flags and search variations.
G2G_AI_Cache.json	Persistent local AI conversions; generated alongside the script.

How to judge results
The most useful metrics are **new .lrc saved**, provider successes, and rejected candidates. “Not found” can mean a genuine absence, a spelling mismatch, or a temporarily unavailable source.

09 Troubleshooting and resources

Issue	What to try
Window closes instantly	Run START_HERE.cmd from PowerShell; then Run_Diagnostics.cmd.
“Module not found”	Let START_HERE.cmd install core dependencies or check your Python interpreter.
G2G model not loading	Run SETUP_GREEK_AI.cmd or manual commands on page 3. Disable AI meanwhile.
AI makes strange Greek words	Review LRC_G2G_AI_Review CSV and add confirmed aliases.
Very slow searches	Test FAST instead of FULL; reduce test size; check provider statistics.
429 / blocked provider	Do not relaunch repeatedly. Provider limits are outside the program’s control.
No lyrics on car radio	Check identical MP3/LRC basenames, timestamps and text encoding; test a small USB sample.

Further information

G2G model: github.com/nlpaueb/gr-nlp-toolkit
Python: python.org/downloads
LRCLIB: lrclib.net
SyncedLyrics: github.com/moehmeni/syncedlyrics

Distribution and responsible use
This public edition contains no private MP3 list, no audio/lyric catalog, no user cache and no bundled G2G model weights. Provider availability and rights vary. Respect each service’s terms, rate limits and applicable copyrights.

Want to get started? Extract the ZIP, optionally run **SETUP_GREEK_AI.cmd**, and start **START_HERE.cmd**. Choose 20 tracks first; inspect the resulting files before a large run.

10 Verification, review and search history

This publication-focused update favors **correct matches over high download counts**. A synchronized file that merely resembles your song is not proof of the same recording.

What is required before an automatic .lrc save?

Requirement	New rule
Artist and title	The source must return resolved artist AND title metadata. Both must meet similarity thresholds.
Recording version	Radio, extended, named remix, instrumental and other identifiable variants cannot be interchanged.
Recording duration	The source must identify a duration and its difference from the MP3 must be at most 3 seconds.
Timed lyrics	At least three timestamps with plausible timing; offset and duplicate timestamps are normalized.

What happens to uncertain lyrics?

Lyrics returned from search-only providers without reliable recording identity are **NOT installed** next to the MP3. Where usable timed text exists, it is saved as **.candidate.txt** under **LRC_Review_Only_v2_1_4** for manual inspection. The Review CSV points to these files. A review file is never assumed to be correct just because it contains timestamps.

New optional history and report archive

File or feature	Meaning
LRC_Completed_Search_History.jsonl	Seven-day cache skips only true empty NO_MATCH searches. REVIEW_AVAILABLE is recorded but never used to skip future queries.
Startup prompt [Y/n]	Choose N to force fresh provider queries. Changed aliases or new versions automatically use different keys.
LRC_Run_Reports / Run_...	Time-stamped copies of results from each completed or interrupted run; familiar main report filenames remain.

Publication checklist
Before public distribution, test on a Windows computer with a small collection, check one standard song plus radio/extended/remix variants, verify review files are not auto-installed, and confirm provider/API terms. Offline tests cannot guarantee live service compatibility.

11 What changed in v2.1.4

This **reliability update** retains the existing command-line workflow and 12 lyrics routes. It fixes how uncertain candidates, repeated searches and API cooldowns are classified.

Three distinct outcomes

Outcome	What it means and what happens
SAVED	A verified recording match was written as a new .lrc file.
REVIEW_AVAILABLE	A source returned candidate timed lyrics, but identity/timing is uncertain; no automatic sidecar.
NO_MATCH	A provider completed its search with no usable lyrics or review candidate. Only this result is eligible for the seven-day skip cache.

New public configuration: LRCLIB identity

LRCLIB asks applications to identify themselves with a real **project homepage URL or contact email**. Edit the included **LRCLIB_CONTACT.txt** and enter one valid HTTPS URL or email below its comments. No sample or invented URL should be used. If left blank, this provider is disabled; the other search sources work normally. The contact value is sent only to LRCLIB.

```
LRCLIB_CONTACT.txt
https://github.com/YOUR-ACCOUNT/YOUR-PROJECT
```

The address above illustrates the format only: replace it with your actual published URL or your real contact email.

429 handling and persistent cooldown

A server can return **429** with **Retry-After**. v2.1.4 records the cooldown as **LRC_Provider_Cooldowns.json** in the chosen music folder and skips that provider during the active period, even after restarting. It does not turn a 429 into a Round 2/3 song retry.

New reports

Output	Use
LRC_Review_Needed.txt	Easy list of songs with candidate lyrics to inspect manually.
Provider Details CSV	Separate SAVED, REVIEW_AVAILABLE, NO_MATCH, CACHED_NO_MATCH, errors and seconds per call.
Source Statistics CSV	Totals for verified saves, review candidates, no-matches and request timings.